

Final exam solutions

1. *Multiple choice (20 points)*. Please mark each listed statement as true ($\checkmark$) or false ($\times$). For each answer that is correctly marked as either true or false you will get 1 point, resulting in 4 points per question.

- (a) Let $Y \in \{1, \dots, K\}$ be a class variable and $X = (X_1, \dots, X_d)$ a feature vector. A maximum a posteriori Bayesian classifier predicts $\hat{y}(x) = \operatorname{argmax}_y \mathbf{P}(Y = y \mid X = x)$. Which of the following statements are true?

- ☐ Changing the class prior $\mathbf{P}(Y = y)$ can change $\hat{y}(x)$ even if every class-conditional density $p(x \mid y)$ remains fixed.
- ☐ The naive Bayes assumption is the marginal factorization

$$p(x_1, \dots, x_d) = \prod_{i=1}^d p(x_i).$$

- ☐ Gaussian naive Bayes models can be directly applied to continuous features without discretization.
- ☐ In a tree-augmented naive Bayes classifier, the feature subgraph may contain an arbitrary number of feature parents for each X_i , so that $p(x \mid y) = \prod_{i=1}^d p(x_i \mid y, \mathbf{pa}(X_i))$ without any restriction on $|\mathbf{pa}(X_i)|$.

- (b) Consider a finite-state Markov chain with transition matrix P , and a hidden Markov model with hidden states $X_{1:T}$ and observations $Y_{1:T} = y_{1:T}$. Which of the following statements are true?

- ☐ The forward algorithm computes $p(y_{1:T}) = \sum_{x_{1:T}} p(x_{1:T}, y_{1:T})$ recursively.
- ☐ In general,

$$\operatorname{argmax}_{x_{1:T}} p(x_{1:T} \mid y_{1:T})$$

need not equal the sequence formed by $\operatorname{argmax}_{x_t} p(x_t \mid y_{1:T})$ for each t .

- ☐ Baum-Welch parameter learning is an instance of the expectation-maximization algorithm.
- ☐ For every transition matrix P , there exists a unique probability vector π satisfying the stationary distribution condition $P^T \pi = \pi$.

- (c) Let $G = (V, E)$ be a Bayesian network. Which of the following statements are true about inference and learning in G ?

- ☐ If $X_A \rightarrow X_C \leftarrow X_B$ is a collider, conditioning on X_C can make $X_A \not\perp\!\!\!\perp X_B \mid X_C$.

- ☐ Consider the variable elimination algorithm for the inference of a fixed query $p(X_Q \mid X_O = x_O)$, changing the variable elimination order can change the sizes of intermediate factors but not the exact result.
 - ☐ If the underlying undirected graph of G contains at most one path between any two nodes, belief propagation computes exact node marginals.
 - ☐ Consider the parameter learning in Bayesian networks. Laplace smoothing increases the number of zero entries in an estimate $\hat{p}(X_i \mid \mathbf{pa}(X_i))$.
- (d) Suppose we want to estimate $\mu = \mathbf{E} f(X)$ with respect to $X \sim p$ for a target density $p(x) = \tilde{p}(x)/Z$, where the normalizing constant Z is unknown. We draw independent samples $x^{(1)}, \dots, x^{(N)}$ from a proposal density q and use the weights $w^{(i)} = \tilde{p}(x^{(i)})/q(x^{(i)})$ and $\bar{w}^{(i)} = w^{(i)} / \sum_{j=1}^N w^{(j)}$. Which of the following statements are true?
- ☐ A valid importance sampling proposal must satisfy $q(x) \geq p(x)$ for every x , and $q(x) > 0$ whenever $p(x) > 0$.
 - ☐ Replacing every weight $w^{(i)}$ by $cw^{(i)}$ for a constant $c > 0$ changes the value of the self-normalized estimator.
 - ☐ For every finite N , the estimator

$$\hat{\mu}_N = \sum_{i=1}^N \bar{w}^{(i)} f(x^{(i)})$$

satisfies $\mathbf{E} \hat{\mu}_N = \mu$.

- ☐ If $q = p$, then $\bar{w}^{(i)} = 1/N$ for all i , and $\hat{\mu}_N$ reduces to the ordinary sample average.
- (e) Consider a finite discounted Markov decision process $(\mathcal{S}, \mathcal{A}, p, r, \gamma)$ with $0 \leq \gamma < 1$ and a policy $\pi(a \mid s)$. Which of the following statements are true?
- ☐ $V^\pi(s) = \mathbf{E}_\pi(G_0 \mid S_0 = s)$, where $G_0 = \sum_{t=0}^{\infty} \gamma^t R_t$.
 - ☐ A Bellman policy evaluation update for $V^\pi(s)$ takes a maximum over $a \in \mathcal{A}$.
 - ☐ Value iteration converges to the unique fixed point V^* of the Bellman optimality operator.
 - ☐ A maximum entropy control objective contains an expected return term and a term proportional to $\mathcal{H}(\pi(\cdot \mid s))$.

Solution.

- (a) $\checkmark \times \checkmark \times$
 - (b) $\checkmark \checkmark \checkmark \times$
 - (c) $\checkmark \checkmark \checkmark \times$
 - (d) $\times \times \times \checkmark$
 - (e) $\checkmark \times \checkmark \checkmark$
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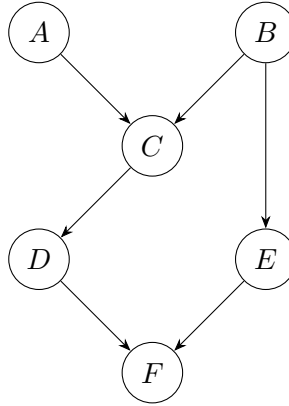
2. *Bayesian networks (20 points).* Consider six discrete random variables, A, B, C, D, E , and F . The parent set of each variable is given by:

- $\text{pa}(A) = \emptyset$,
- $\text{pa}(B) = \emptyset$,
- $\text{pa}(C) = \{A, B\}$,
- $\text{pa}(D) = \{C\}$,
- $\text{pa}(E) = \{B\}$,
- $\text{pa}(F) = \{D, E\}$.

- (a) (6 points.) Draw the *directed acyclic graph* for this Bayesian network. Based on the structure, write down the factorization of the joint probability distribution of all variables $\mathbf{P}(A, B, C, D, E, F)$.
- (b) Using the ideas of *d-separation*, determine whether the following statements are true or false for any distribution compatible with the graph. Justify your answer by identifying the type of path, *e.g.*, chain, fork, or inverted fork (collider).
- i. (2 points.) $(A \perp\!\!\!\perp B)$.
 - ii. (2 points.) $(A \perp\!\!\!\perp B \mid D)$.
 - iii. (2 points.) $(D \perp\!\!\!\perp E \mid B)$.
 - iv. (2 points.) $(D \perp\!\!\!\perp E \mid B, F)$.
- (c) (6 points.) For this Bayesian network, what is the Markov blanket of each variable?

Solution.

(a) The directed acyclic graph for the Bayesian network is as follows:



The joint probability distribution factorizes as

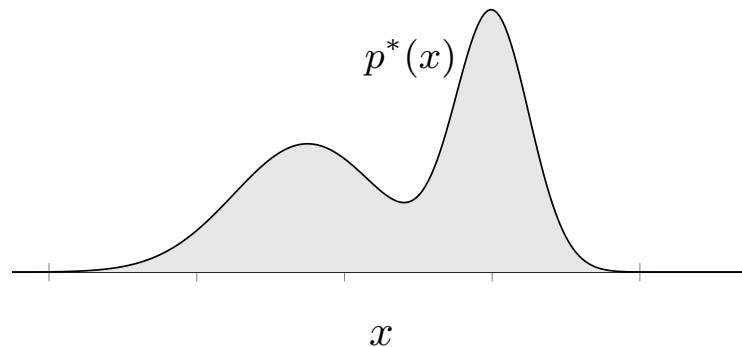
$$\mathbf{P}(A, B, C, D, E, F) = \mathbf{P}(A) \mathbf{P}(B) \mathbf{P}(C \mid A, B) \mathbf{P}(D \mid C) \mathbf{P}(E \mid B) \mathbf{P}(F \mid D, E).$$

- (b) i. True. On the path $A \rightarrow C \leftarrow B$, C is an unobserved collider. The other path, $A \rightarrow C \rightarrow D \rightarrow F \leftarrow E \leftarrow B$, is blocked by the unobserved collider F . Hence, all paths between A and B are blocked.

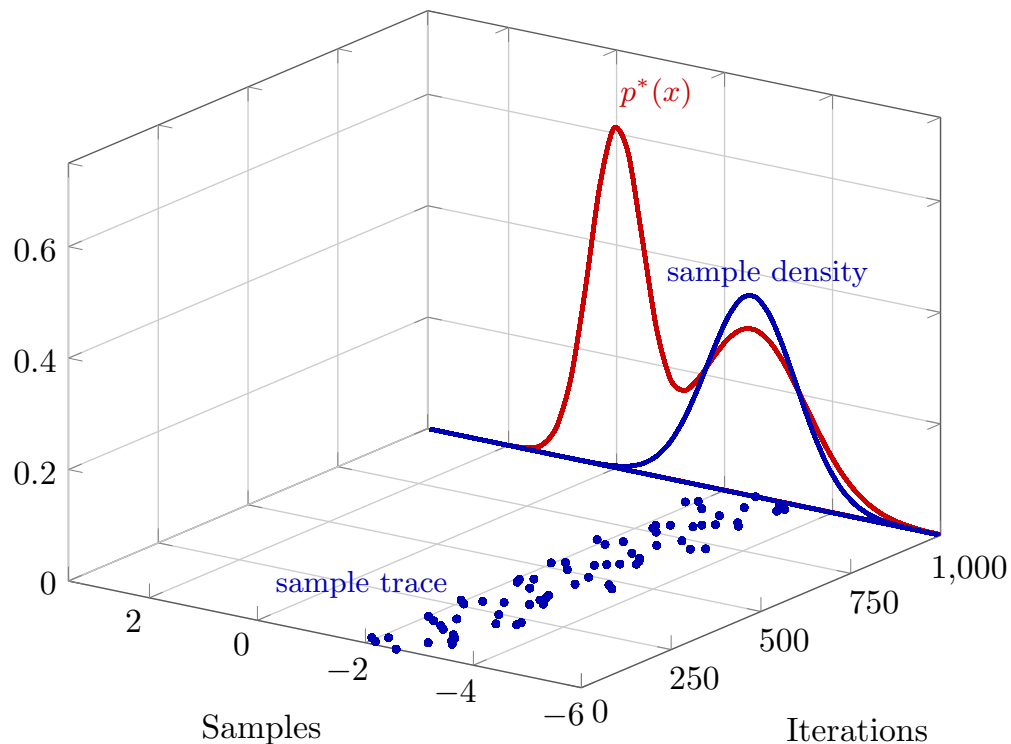
- ii. False. Since D is a descendant of the collider C , conditioning on D opens the path $A \rightarrow C \leftarrow B$. Thus, A and B are not d -separated given D .
 - iii. True. Conditioning on B blocks the path $D \leftarrow C \leftarrow B \rightarrow E$, while the path $D \rightarrow F \leftarrow E$ is blocked by the unobserved collider F . Hence, all paths between D and E are blocked given B .
 - iv. False. Conditioning on the collider F opens the path $D \rightarrow F \leftarrow E$. Therefore, D and E are not d -separated given B and F .
- (c) The Markov blankets are
- $\mathbf{mb}(A) = \{B, C\}$,
 - $\mathbf{mb}(B) = \{A, C, E\}$,
 - $\mathbf{mb}(C) = \{A, B, D\}$,
 - $\mathbf{mb}(D) = \{C, E, F\}$,
 - $\mathbf{mb}(E) = \{B, D, F\}$,
 - $\mathbf{mb}(F) = \{D, E\}$.
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3. *Metropolis-Hastings algorithm (10 points).* Suppose we want to sample from a target distribution $p^*(x)$, which is known up to a normalizing constant, *i.e.*, $p^*(x) = \tilde{p}(x)/Z$. Let q be a proposal distribution that is easy to sample from, where $q(x' | x)$ denotes the probability of proposing x' given the current state x . We assume that q is asymmetric, *i.e.*, $q(x' | x) \neq q(x | x')$ in general.

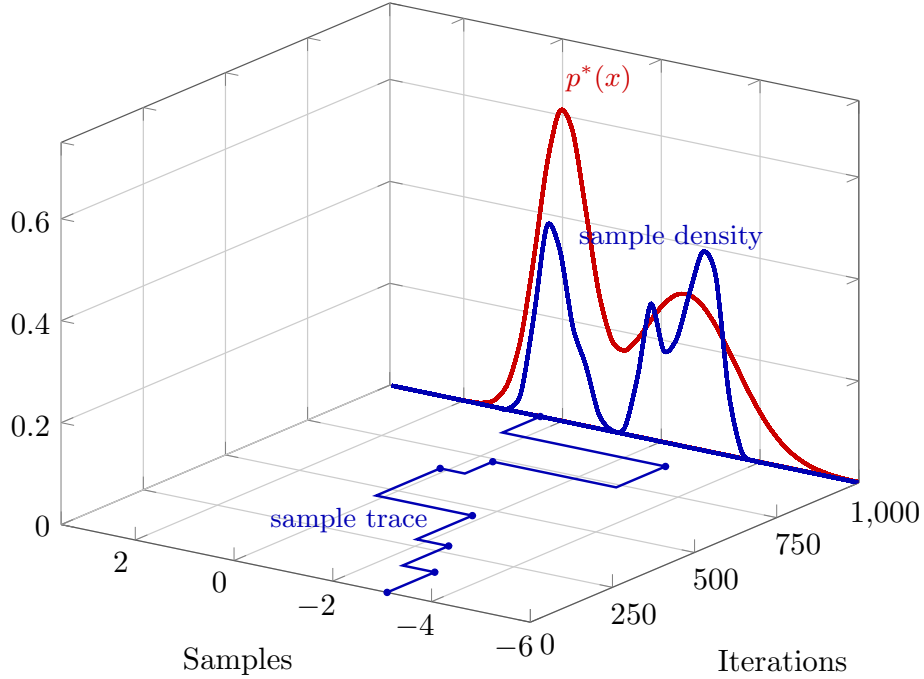
- (a) (5 points.) Describe the Metropolis-Hastings algorithm for generating samples from $p^*(x)$. Note that only the unnormalized target density $\tilde{p}(x)$ is known and can be evaluated.
- (b) Suppose the target distribution is a mixture of two Gaussian distributions shown in the figure below, and the proposal distribution is a Gaussian distribution centered at the current state x with a fixed variance σ^2 , *i.e.*, $q(x' | x) = \mathcal{N}(x, \sigma^2)$.



- i. (3 points.) Consider the case when you see the following sample distribution after running the Metropolis-Hastings algorithm for a long time.



- What is the likely reason for this sample distribution? How can you fix it?
- ii. (2 points.) Do the same reasoning for the following sample distribution.



Solution.

- (a) Choose an initial state x_0 in the support of the target distribution. For $t = 0, 1, \dots$, perform the following steps:

- i. Draw a proposal

$$x' \sim q(\cdot | x_t).$$

- ii. Compute the Metropolis-Hastings acceptance probability

$$A(x' | x_t) = \min \left\{ 1, \frac{\tilde{p}(x')q(x_t | x')}{\tilde{p}(x_t)q(x' | x_t)} \right\}.$$

- iii. Draw $u \sim \mathcal{U}(0, 1)$ and set

$$x_{t+1} = \begin{cases} x', & u \leq A(x' | x_t), \\ x_t, & u > A(x' | x_t). \end{cases}$$

- (b) i. The trace and empirical density indicate that the chain is trapped in the broader left mode and has not mixed between the two modes. The likely cause is that σ^2 is too small relative to the separation between the modes: almost all proposals remain near the current mode, while proposals crossing the low-density region are extremely rare. Increasing σ^2 makes direct moves between the modes more likely.
- ii. The long stretches of identical samples show that most proposals are rejected. The likely cause is that σ^2 is too large: proposed states often lie in regions where the target density is very small and hence have a low acceptance probability. Decreasing σ^2 increases the acceptance rate and allows the chain to move more frequently.
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4. *Linear output hidden Markov models (10 points).* Suppose we are given a dataset

$$(x(t), y(t)), \quad t = 1, \dots, m,$$

where $x(t) \in \mathbf{R}^n$ is the feature vector and $y(t) \in \mathbf{R}$ is the corresponding output, observed from a linear output hidden Markov model with K hidden states. It is assumed that the output $y(t)$ at each time step is generated from a linear measurement model, given by

$$y(t) = \theta_{z(t)}^T x(t) + \epsilon(t),$$

where $z(t) \in \{1, \dots, K\}$ is the hidden state at time t , $\theta_{z(t)} \in \mathbf{R}^n$ is the coefficient vector associated with state $z(t)$, and $\epsilon(t)$ is standard Gaussian noise.

We would like to estimate the parameters of the model, including the state transition probabilities P and the linear measurement coefficients $\theta_1, \dots, \theta_K$, based on the observed dataset. We consider a two-step procedure, which first estimates the coefficients θ_i together with the hidden states $z(t)$, so that in the second step the state transition probabilities P can be estimated based on the inferred hidden states.

- (a) The first step consists in minimizing the total squared error between the observed outputs and the outputs predicted by the linear output hidden Markov model. For simplicity of notation, we may transform the hidden state labels into one-hot vectors, *i.e.*, let $z(t) \in \{e_1, \dots, e_K\} \subseteq \mathbf{R}^K$, where e_i is the i -th standard basis vector in $\mathbf{R}^K$, which has 1 in the i -th entry and 0 elsewhere. Then we can formulate the first step of the procedure as an optimization problem, which is given by

$$\begin{aligned} & \text{minimize} && J(\theta_1, \dots, \theta_K, z(1), \dots, z(m)) \\ & \text{subject to} && z(t) \in \{e_1, \dots, e_K\}, \quad t = 1, \dots, m \end{aligned} \tag{1}$$

where the optimization variables are $\theta_1, \dots, \theta_K \in \mathbf{R}^n$ and $z(1), \dots, z(m) \in \mathbf{R}^K$.

- i. (3 points.) Please give an explicit expression of the objective J in the problem (1), as a function of the linear coefficients θ_i and the one-hot vectors $z(t)$.

In your objective, what is the role of the one-hot vector $z(t)$?

Hint. You may find it useful to write $z_i(t)$ for the i -th component of $z(t)$, so that $z_i(t) = 1$ if $z(t) = e_i$ and $z_i(t) = 0$ otherwise.

- ii. (2 points.) Based on your expression for J , please describe an algorithm for approximately solving the optimization problem (1). In particular, you should explain what is the objective and the variables of each subproblem in your formulation, and how they are updated.

Hint. You might get some inspiration from the *expectation-maximization* algorithm or the *k-means clustering*.

- (b) (5 points.) Now suppose we have obtained an estimate of the hidden states $\tilde{z}(t) \in \{1, \dots, K\}$ for all $t = 1, \dots, m$. Explain how to estimate the state transition probabilities P based on the inferred hidden states. You can either express your answer in words or mathematically.

Solution.

- (a) i. Writing $z_i(t)$ for the i -th component of $z(t)$, the objective is

$$J(\theta_1, \dots, \theta_K, z(1), \dots, z(m)) = \sum_{t=1}^m \sum_{i=1}^K z_i(t) (y(t) - \theta_i^T x(t))^2.$$

Equivalently, each summand is the inner product of $z(t)$ with the vector of squared prediction errors for the K models. Since $z(t)$ is one-hot, it selects the error of the model corresponding to the hidden state at time t .

- ii. One may use alternating minimization, starting from initial values of $\theta_1, \dots, \theta_K$ and repeating the following two steps until the assignments (or the objective) no longer change:
- With the coefficients fixed, assign each observation to the model with the smallest squared prediction error:

$$z(t) := e_{i^*}, \quad i^* \in \operatorname{argmin}_{i=1, \dots, K} (y(t) - \theta_i^T x(t))^2.$$

- With the assignments fixed, update each coefficient vector by a least-squares regression on the observations assigned to state i :

$$\theta_i := \operatorname{argmin}_{\theta \in \mathbf{R}^n} \sum_{\{t | z(t) = e_i\}} (y(t) - \theta^T x(t))^2, \quad i = 1, \dots, K.$$

Each step minimizes J over one group of variables while keeping the other group fixed, so the objective cannot increase, although the procedure need not find a global optimum.

- (b) Count the inferred transitions from each state to every other state. In particular, let

$$N_{ij} = \sum_{t=1}^{m-1} I(\tilde{z}(t) = i, \tilde{z}(t+1) = j),$$

where $I(\cdot)$ is the indicator function. Then N_{ij} is the number of observed transitions from state i to state j . The maximum likelihood estimate is obtained by normalizing each row of the count matrix:

$$\hat{P}_{ij} = \frac{N_{ij}}{\sum_{k=1}^K N_{ik}}.$$

Thus, $\hat{P}_{ij}$ is the fraction of observed transitions out of state i that go to state j . (If no transition out of a state is observed, its row cannot be estimated without an additional convention or prior.)
